x_1 x_2 x_3 x_4 x_5 x_6

$\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}2\\1\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}5\\4\end{pmatrix}\begin{pmatrix}6\\5\end{pmatrix}\begin{pmatrix}6\\6\end{pmatrix}$

C_1 C_1 C_1 C_2 C_2 C_2

$\begin{vmatrix}4\\-2\\-2\end{vmatrix}$ *Initial W* **REPEAT using Step η = 1**

C_1 positive C_2 negative Zero is unaccepted

Check classes of Z_1 $\begin{vmatrix}0\\0\end{vmatrix}$ Z_2 $\begin{vmatrix}7\\6\end{vmatrix}$ Z_3 $\begin{vmatrix}3\\2\end{vmatrix}$

Ugmented Vectors (from Top)

x_1 $\begin{vmatrix}1\\1\\1\end{vmatrix}$ x_2 $\begin{vmatrix}1\\2\\1\end{vmatrix}$ x_3 $\begin{vmatrix}1\\1\\2\end{vmatrix}$ x_4 $\begin{vmatrix}1\\5\\4\end{vmatrix}$ x_5 $\begin{vmatrix}1\\6\\5\end{vmatrix}$ x_6 $\begin{vmatrix}1\\6\\6\end{vmatrix}$

C_1 C_1 C_1 C_2 C_2 C_2

Iteration 2

No change in W
Stop Iterations

$d(x_1) = \begin{matrix}1 & 1 & 1\end{matrix}\begin{vmatrix}5\\-1\\-1\end{vmatrix} = 3$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix}5\\-1\\-1\end{vmatrix}$

$d(x_2) = \begin{matrix}1 & 2 & 1\end{matrix}\begin{vmatrix}5\\-1\\-1\end{vmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix}5\\-1\\-1\end{vmatrix}$

$d(x_3) = \begin{matrix}1 & 1 & 2\end{matrix}\begin{vmatrix}5\\-1\\-1\end{vmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix}5\\-1\\-1\end{vmatrix}$

$d(x_4) = \begin{matrix}1 & 5 & 4\end{matrix}\begin{vmatrix}5\\-1\\-1\end{vmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix}5\\-1\\-1\end{vmatrix}$

$d(x_5) = \begin{matrix}1 & 6 & 5\end{matrix}\begin{vmatrix}5\\-1\\-1\end{vmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix}5\\-1\\-1\end{vmatrix}$

$d(x_6) = \begin{matrix}1 & 6 & 6\end{matrix}\begin{vmatrix}5\\-1\\-1\end{vmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{vmatrix}5\\-1\\-1\end{vmatrix}$